MILITARY SPECIFICATION

COMPUTER SET, NAVIGATIONAL

AN/ASN-39(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall compute and provide electrical outputs of great circle bearing (relative to heading) to a target or base as selected, ground track (relative to heading) and great circle distance to a target or a base as selected. The equipment shall also compute and display the present position of the aircraft in latitude and longitude co-ordinates.

1.2 Classification - The Computer Set, Navigational shall consist of the following items:

Item	Type Designation	Appl. Para.
Computer-Control	CP-624(*)/ASN-39	3.4.1
Amplifier-Computer	AM-3049(*)/ASN-39	3.4.2

1.3 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

(a) Computer Set, Bomb Release, Loft AN/AJB-3

(b) Central Air Data Computer (CADC) System

(c) Indicator - ID-663A/U

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

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3.2.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 25.0 pounds.

3.2.2 Reliability

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.2.2.3 Reliability In Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 100 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.2.4 Time Totalizing Meter - The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793 and MS28053. The meter shall be included in the Amplifier-Computer.

3.2.3 Cabling and Connections

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

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3.2.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage 115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC	

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.2.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Figure 1.

3.2.10.2 Temperature - The equipment shall meet the requirements for Class 2 environmental service conditions except that at temperatures between -35° C and -55° C the equipment accuracy shall not exceed two times the accuracy specified in 3.3.2.

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3.2.10.3 Acoustic Noise - No malfunction of the equipment shall occur when subjected to a jet engine noise field with an intensity of 120 ± 5 db relative to .0002 dyne per square centimeter. This noise field contains all frequencies between 20 and 10,000 cps, with relative amplitude as shown on Figure 2.

3.2.10.4 Equipment Maneuver Load Factors - Limit Load - The equipment, when normally mounted, shall not suffer damage or subsequently fail to provide normal performance when subjected to the following limit load factors for protracted periods.

Vertical Down	Vertical Up	Horizontal Major and Minor Axis
6.7g	3.3g	±6.1g

3.2.11 Warm-up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 6 minutes at extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 100 VA

(2) AC Power (Three-Phase) 115/200V, Category "B", 20 VA

(3) DC Power 28V, Category "B", 0.5 Amp

3.2.12.1 Degraded Performance - Degraded performance will be permitted for all voltages transients from 0 to 0.5 sec. duration as shown on Figure 2 of MIL-STD-704 for Normal Electric System Operation except performance shall not be affected by short pulses of less than 50 microseconds. Operation shall return to normal when the 0 to 5 sec. transient expires.

3.2.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all parts of the Navigation Computer Set except to the latitude and longitude integrator channels.

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3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The equipment shall continuously compute present position in latitude and longitude, bearing and distance to a target or a base (as selected) and ground track. The present position shall be displayed on the Computer-Control; the bearing, distance and ground track shall be supplied as electrical outputs to be displayed on the associated equipment indicators (BDHI). Ground track and bearing shall be relative to heading. The equipment shall accept automatic inputs of True Airspeed from the associated Air Data Computer and Heading from the associated heading reference. Manual inputs of Wind Direction, Wind Velocity, Magnetic Variation, Present Position, and Target Position shall be set in at the Computer-Control. The navigation computer shall compute a rhumb line range and bearing to target over the range of 0 to 200 miles and shall compute a great circle range and bearing over the range of 200 to 2000 miles.

3.3.2 Electrical Input Signals - The navigation computer set shall accept the following electrical input signals:

3.3.2.1 True Air Speed - The navigation computer set shall accept an input signal representing true air speed over the range of 150 to 1500 knots. The input signal shall be a voltage from a 5000 ohm linear potentiometer located in the associated Central Air Data Computer. The 5000 ohm linear potentiometer shall be excited with 28 volts, 400 cps by the navigation computer set.

3.3.2.2 Magnetic Heading - The navigation computer set shall accept an input of magnetic heading. The magnetic heading input shall be in synchro from (Eclipse-Pioneer AY-20 /___/-4-B, or equivalent) at a scale factor of 1°/1°.

3.3.3 Manual Inputs - The navigation computer set shall accept the following manual inputs:

3.3.3.1 Present Position: The navigation computer set shall accept a manual input of present position in latitude and longitude coordinates. The input shall be a cyclometer counter over the longitude range of 180° 00' E through 0° to 180° 00' W in 1 minute increments and over the latitude range of 90° 00' N through 0° to 90° 00' S in 1 minute increments.

3.3.3.2 Target Position - The navigation computer set shall accept a manual input of target position in latitude and longitude coordinates. The input shall be on a cyclometer counter over the longitude range of 180° W through 0° to 180° 00' E in 1 minute increments and over the latitude range of 90° 00' S through 0° to 90° 00' N in one minute increments.

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3.3.3.3 Magnetic Variation - The Navigation Computer Set shall accept a manual input of magnetic variation on a cyclometer counter over the range of 180° W through 0° to 180° E in one degree increments.

3.3.3.4 Wind Speed - The Navigation Computer Set shall accept a manual input of wind speed in knots on a cyclometer counter over the range of 0 to 200 knots in 1 knot increments.

3.3.3.5 Wind Direction - The Navigation Computer Set shall accept a manual input of wind direction on a cyclometer counter over the range of 0° to 360° in one degree increments. The wind direction shall be designated as the "from" direction.

3.3.4 Electrical Output Signals - The Navigation Computer Set shall provide electrical output signals as shown in Table 1.

Table 1

Electrical Output Signals

Output	Form	Range	Scale Factor
Ground Track	Synchro Eclipse-Pioneer AY-193-T1 or equivalent	0° to 360°	1°/1°
Bearing to Target	Synchro, Eclipse-Pioneer AY-193-T1 or equivalent	0° to 360°	1°/1°
Distance to Target	Three Synchros Eclipse-Pioneer AY-503F-21-A3 or equivalent and close contacts	0 to 1999	Contacts closed for 1000 mile flag Three synchros one each as follows: Units 0 to 9 Tens 10 to 99 Hundreds 100 to 999

3.3.5 Visual Outputs - The Navigation Computer Set shall provide a visual output present position in latitude and longitude coordinates. The present position output shall be displayed on the same cyclometers counters used for present position inputs as defined in 3.3.3.1. The present position output shall be over the longitude range of 180° 00' E in one minute increments and over the latitude range of 90° 00' N through 0° to 90° 00' S in one minute increments.

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3.3.6 Accuracy - The accuracy specified herein shall be maintained with no adjustments when the Computer Set is connected to one BDHI and one HSI when provided with perfect inputs.

3.3.6.1 Present Position - Present Position Latitude and Longitude errors shall not exceed 1 mile for each 12 minutes of elapsed time after start of problem or P times distance traveled, whichever is greater, where P is defined as follows:

P = 1% latitude computation
P = 1% for longitude computation from the equator to 50° North or 50° South
P = 1-1/2% for longitude computation for 50° - 60° North or South Latitude
P = 2-1/2% for longitude from 60° - 72° North or South latitude

In latitude or longitude, 20% of the test points may exceed these tolerances by 50%.

3.3.6.2 Ground Track Angle - Accuracy of the ground track angle being transmitted to the BDHI shall be as follows:

(a) ± 2° for ground speeds between 150 and 1700 knots.

3.3.6.3 Range and Bearing

(a) Range - The range error for 80% of all test points shall not exceed 5 miles plus 1% of the theoretical range. For the remaining 20% of the test points, these errors shall not exceed 7-1/2 miles plus 1.5 % of theoretical range.

(b) Bearing - The bearing error for 80% of all test points shall not exceed the amount which is defined by a line passing through present position and a point 5 miles to either side of the target or base (as selected or 2° deviation from a line passing through the target or base, whichever tolerance is the larger. For the balance of the test points, the tolerance may be increased by 50%.

3.3.6.4 Accuracy in Degraded Performance - At temperatures between -35°C and -50°C, the specified accuracy shall not exceed two times the values listed in 3.3.6.1, 3.3.6.2 and 3.3.6.3.

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3.4 Detail Requirements

3.4.1 Computer-Control CP-624(*)/ASN-39 - The Computer-Control shall meet the following requirements:

3.4.1.1 Function - The Computer-Control shall be the unit into which the navigation problem is manually inserted. It shall display the Present Position of the carrying aircraft in Latitude and Longitude, the Target or Base in Latitude and Longitude, the Wind Direction in degrees, the Wind Velocity in Knots, and the Variation in degrees. Manual controls shall be provided for setting these indications on cyclometer type counters. A Mode Selector control shall be provided to determine operational modes of OFF, TARGET, BASE, STANDBY, and RESET. This unit shall operate with the Amplifier-Computer, utilizing these manual inputs and automatic inputs to the system of True Airspeed and Heading to provide the over-all requirements.

3.4.1.2 Form Factor - The Computer-Control shall be a console mounted unit in accordance with Specification MIL-C-6781, and shall conform to the dimensions shown on Figure 3.

3.4.1.3 Weight - The weight of the Computer-Control shall not exceed 10.25 lbs.

3.4.1.4 Contents - The Computer-Control shall contain the following major assemblies and circuit functions:

(a) True Airspeed Follow Up - This section shall accept the input from the true airspeed potentiometer in the CADC and by means of a servo system it shall position the shaft of a linear synchro to provide a voltage representing the true airspeed.

(b) Compass Follow Up - This section shall accept the synchro signal from the compass and by means of a servo system it shall position the shaft of a synchro resolver for the true heading computation.

(c) Secant Corrector - This section shall provide correction to the E-W distance required due to convergence of the meridians.

(d) Wind Velocity - This section shall position the shaft of a linear synchro in accordance with the wind velocity counter indication on the panel.

(e) Wind Direction - This section shall position the shaft of a synchro resolver to a position representing the wind direction as set in on the counter on the panel.

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(f) Variation Section - This section shall provide a shaft position in accordance with the counter setting on the panel and this shaft position shall combine with the compass heading shaft position to give a shaft position to the heading resolver to represent true heading.

(g) Great Circle Computation Section - This section shall combine the set-in target (or base) position with the present position setting and perform computation to develop synchro outputs from which the range and bearing can be computed.

(h) Integrator - This section shall utilize the voltages developed in the true heading resolver which represent the velocity in a north-south and an east-west direction as developed from ground speed and true heading, and integrates these velocities with time to produce shaft motions of north south and east west distance. These distances shall then be converted into latitude and longitude to be presented on the present position counters.

(i) The Computer-Control shall provide synchro outputs to the Amplifier Computer of north-south and east-west velocity, true heading, and great circle bearing and range functions. A separate synchro output of true heading shall be provided for use with external equ equipment.

3.4.1.5 Controls - Controls shall be provided for setting each of the displays on the control panel. A detented rotary control shall be provided for selection of the modes of operation. The location of the displays and controls shall be as shown on Figure 2.

3.4.1.5.1 Mode Selector Switch - The mode selector switch shall have five positions: OFF, TARGET, BASE, STBY, RESET.

a) OFF: All electric power is removed from the equipment.

b) TARGET - In the "Target" position computations shall be continuously performed so that target range, bearing and present position shall appear on indicators specified herein.

c) BASE - In the "Base" position computations shall be continuously performed so that base range, bearing and present position shall appear on indicators.

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d) STANDBY - In this mode the equipment shall be energized, but inoperative.

e) RESET - The base memory mechanism shall be aligned with the present position setting.

3.4.1.5.2 Wind Direction Control Knob - To manually insert or change (also in flight) the True Wind Direction.

3.4.1.5.3 Wind Velocity Control Knob - To manually insert or change (also in flight) the wind speed.

3.4.1.5.4 Magnetic Variation Control Knob - To manually insert the magnetic variation angle into the equipment.

3.4.1.5.5 - Position Latitude Control Knob - To permit manually inserting or changing the position Latitude or the base Latitude.

3.4.1.5.6 - Position Longitude Control Knob. To manually insert or change the position Longitude or base Longitude.

3.4.1.5.7 - Target Position Latitude Control Knob - To manually insert or change (also in flight) the destination Latitude position.

3.4.1.5.8 - Target Position Longitude Control Knob - To manually insert or change (also in flight) the destination Longitude position.

3.4.1.6 Indications: Indications shall be provided as follows

3.4.1.6.1 Variation - Magnetic Variation from 0° to 360° shall be displayed on a three drum cyclometer counter. The direction of magnetic variation shall be indicated by display of the words EAST or WEST.

3.4.1.6.2 Wind From - The true wind FROM direction shall be displayed on a three drum cyclometer counter.

3.4.1.6.3 Wind Knots - The magnitude of the wind speed in knots shall be displayed on a three drum cyclometer counter.

3.4.1.6.4 Position Latitude - The position latitude shall be displayed in degrees and minutes north and south.

3.4.1.6.5 Position Longitude - The position longitude shall be displayed in degrees and minutes, east and west.

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3.4.1.6.6 Target Latitude - The target latitude shall be displayed in degrees and minutes, north and south.

3.4.1.6.7 Target Longitude - The target longitude shall be displayed in degrees and minutes, east and west.

3.4.1.7 Zero Range Adjustment - A zero range adjustment shall be provided to minimize range errors.

3.4.1.8 Electrical Connections - Electrical connections shall be provided as shown in Figure 4.

3.4.2 Amplifier-Computer AM-3049(*)/ASN-39

3.4.2.1 Function - The Amplifier-Computer shall provide the routing for the electrical connections and the power supply for the AN/ASN-39 equipment. In addition it shall accept the outputs from the Computer-Control and from these it shall compute and provide synchro outputs representing Ground Track relative to heading, Bearing relative to heading, Range in miles to Target (or Base, as selected on the control panel) in units, tens, hundreds, and a 28V DC signal representing the reaching of 1000 miles.

3.4.2.2 Form Factor - The Amplifier-Computer shall conform to the dimensions and requirements shown on Figure 5.

3.4.2.3 Weight - The weight of the Amplifier-Computer shall not exceed 14.8 lbs.

3.4.2.4 Contents - The amplifier computer shall contain the following major assemblies and functions.

3.4.2.4.1 Range Follow-up. - This is a self-contained module and shall accept its input from the output of the Great Circle chain network. The outputs are range information from the present position to a selected base or target position. Three self synchronous autosyns, one each for units, tens, and hundreds transmit an electrical signal to the appropriate indicators. A switch actuates the "thousands" signal.

3.4.2.4.2 Bearing and Ground Track Follow-up Module - Accepts input signal from the computer and positions the bearing and ground track servos. The outputs of each autosyn transmitter supply these bearing and ground track data to appropriate indicators.

3.4.2.4.3 Amplifiers - Five servo amplifiers provide the necessary amplification for the range, bearing and ground track, compass and true air-speed, servos respectively. The fifth amplifier shall drive the Latitude and Longitude integrators alternately. A channel pre-amplifier provides preamplification for some of these networks.

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3.4.2.4.4 - An electronic board consisting of all the filtering network required by the system.

3.4.2.4.5 - Timer-Precision - A direct current precision timer is used as a timing device for integrating distance.

3.4.2.4.6 - Power Supply - Provides electric power for the entire system.

3.4.2.5 Controls - There shall be no controls on the Amplifier-Computer.

3.4.2.6 Electrical Connections - Shall be as shown in Figure 4.

3.5 Design Data - No data is required by the specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2, unless specified in the contract or order.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized represenative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure I from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094. During each cycle the equipment shall be operated during the heat portion of the cycle.

4.4.3.3 Accept-Reject Criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the qualification test plan. Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the sampling test plan.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 6 hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured -

Characteristic	Spec Requirement	Time in cycle that measurement is to be made
Present Position	Refer to Para 3.3.6.1	During the heat portion of the cycle
Range to target	Refer to Para 3.3.6.3(a)	
Bearing to target	Refer to Para 3.3.6.3(b)	

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test.

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - During the period of the tests no preventive maintenance measures may be performed upon the equipment.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear"items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

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6. NOTES

6.1 Intended Use - This equipment is to be used in high performance aircraft as a self contained navigational computer to provide the aircraft's present position, the bearing and range to a preselected target (or base) and the ground track angle.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.5).

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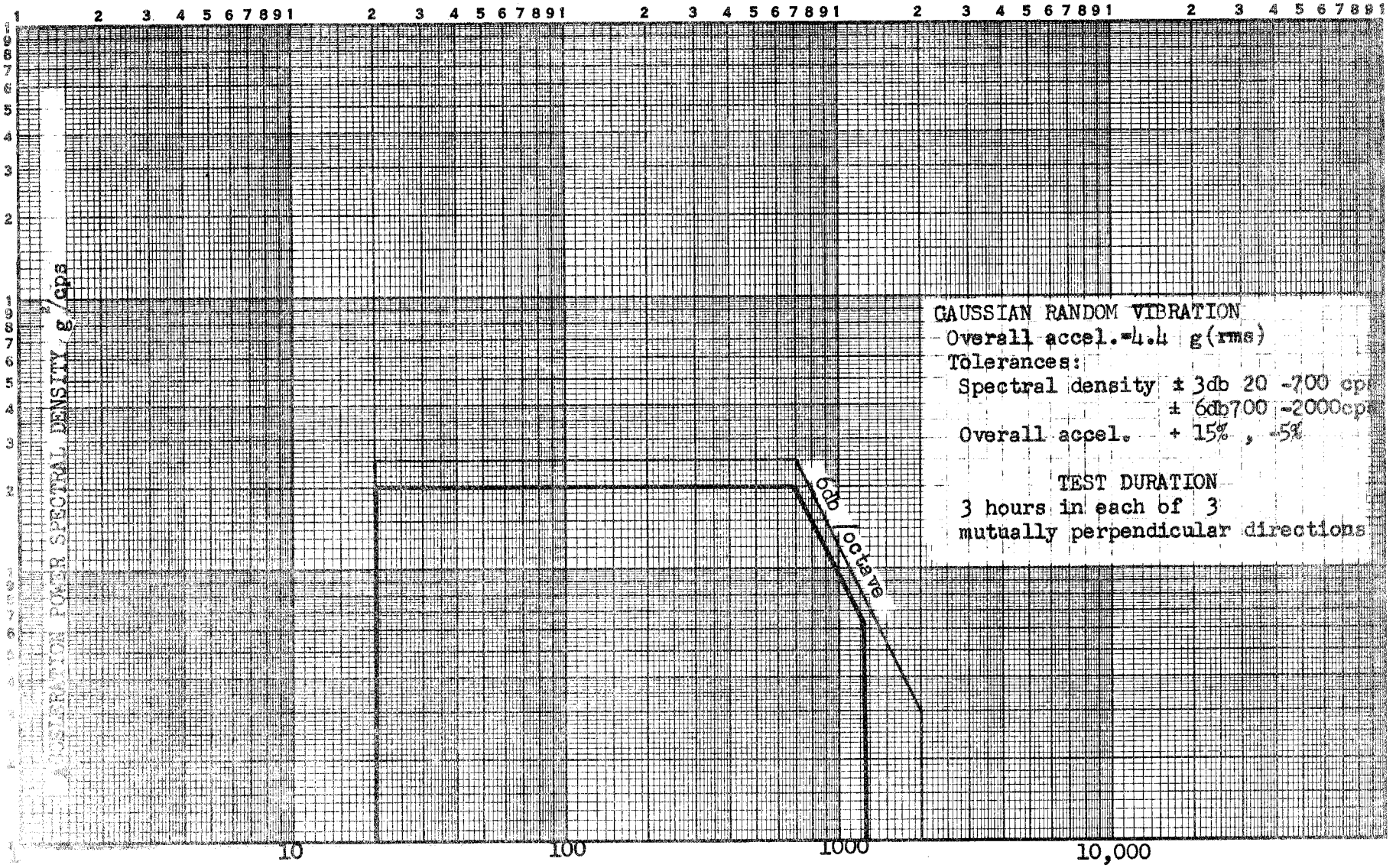
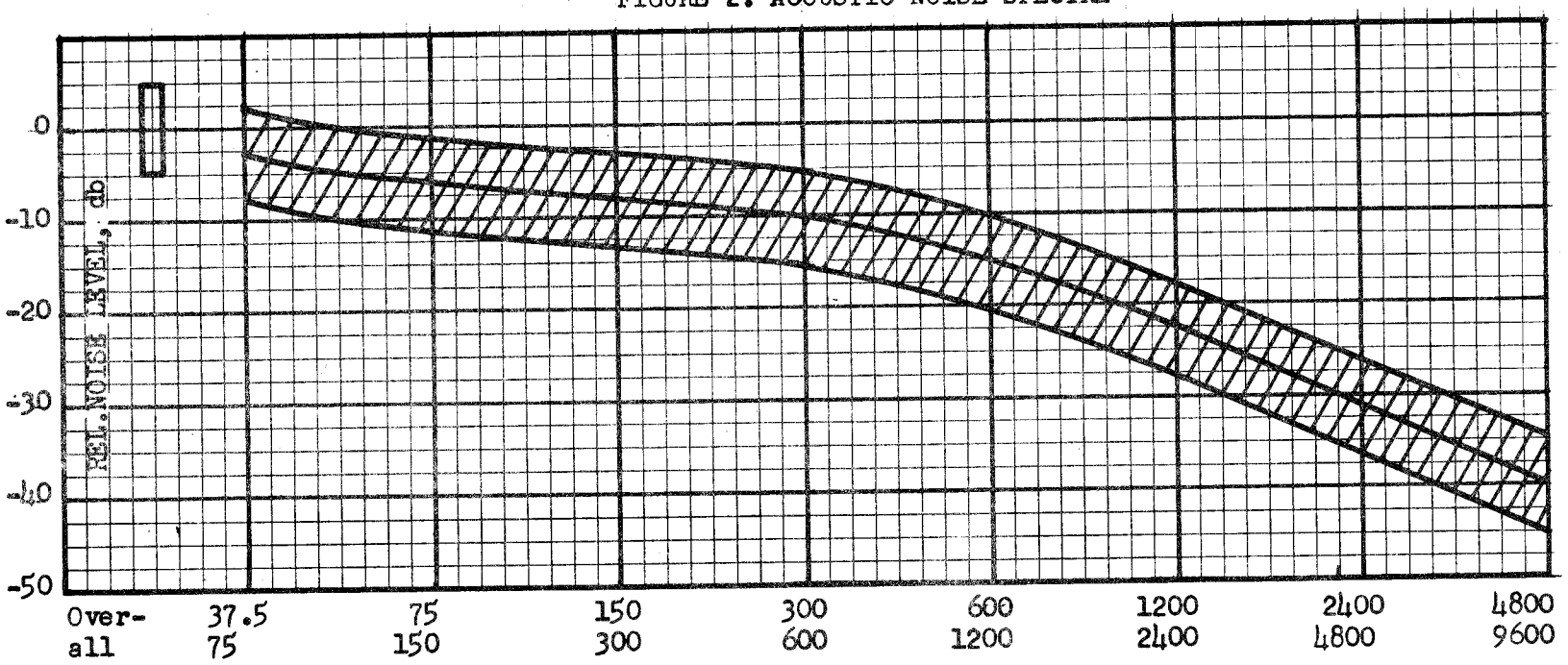


FIGURE I. RANDOM VIBRATION TEST CURVE ENVELOPE

MIL-C-81169(MP)

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FIGURE 2. ACOUSTIC NOISE SPECTRA



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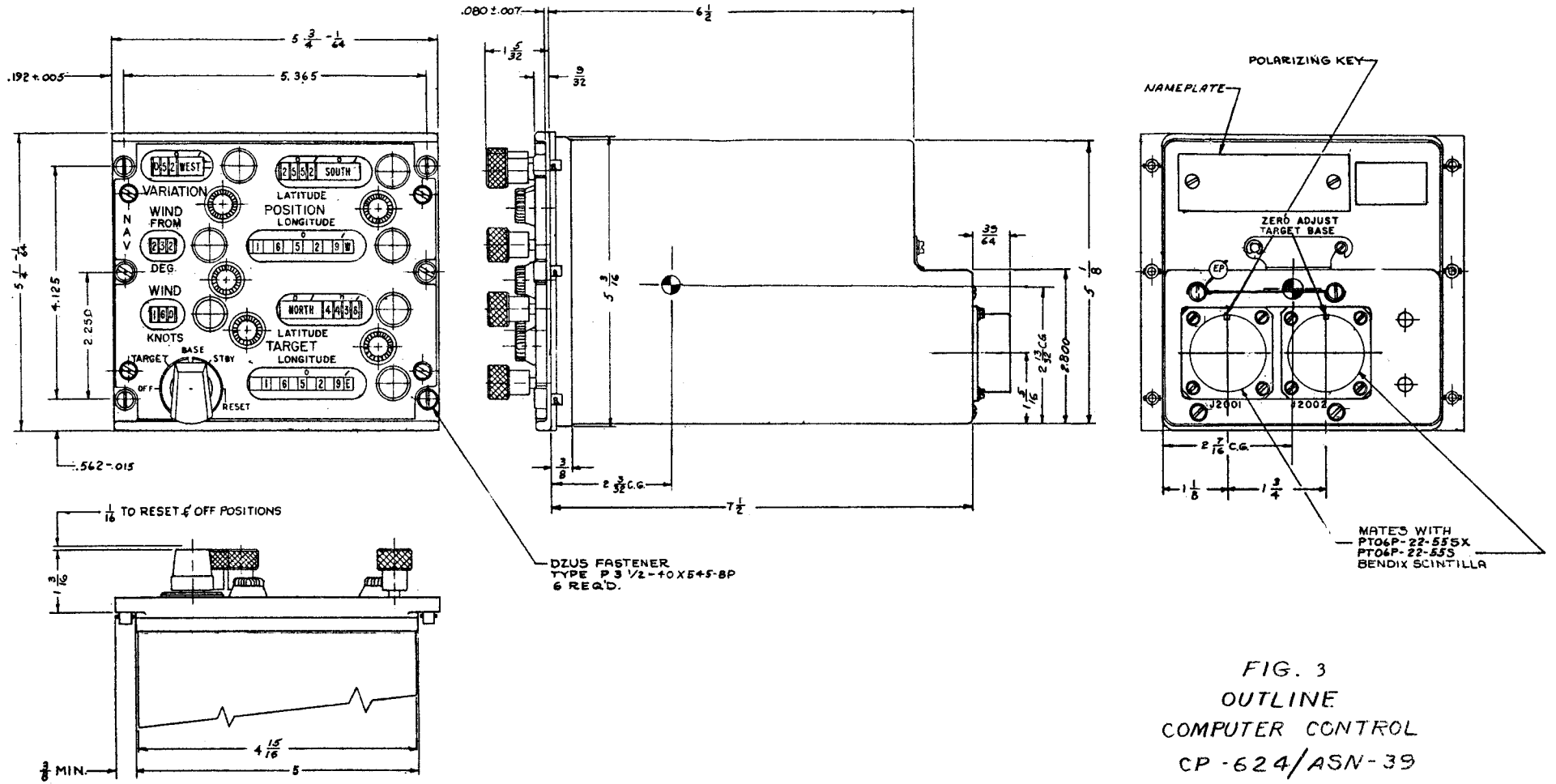


FIG. 3
OUTLINE
COMPUTER CONTROL
CP-624/ASN-39

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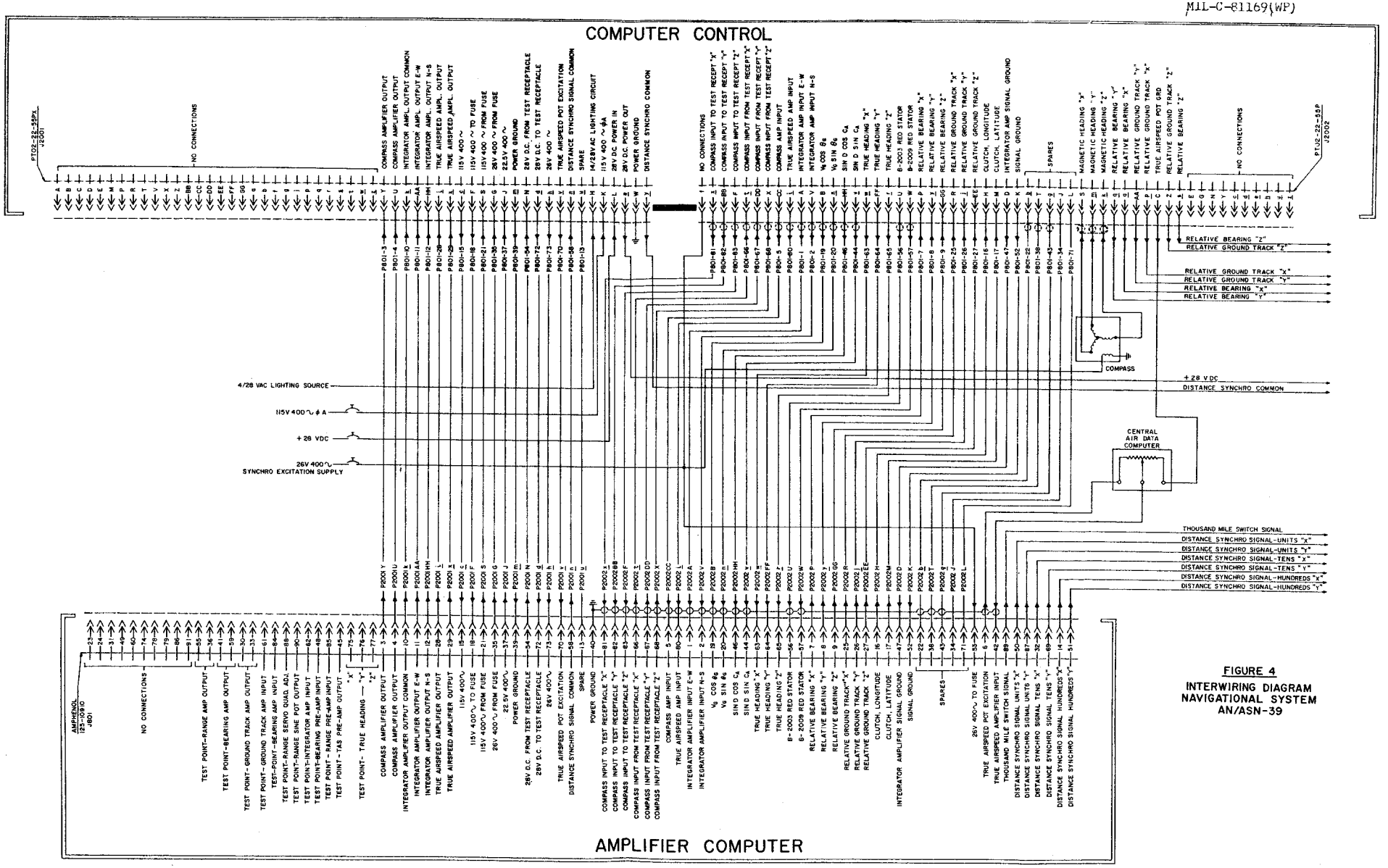


FIGURE 4
INTERWIRING DIAGRAM
NAVIGATIONAL SYSTEM
AN/ASN-39

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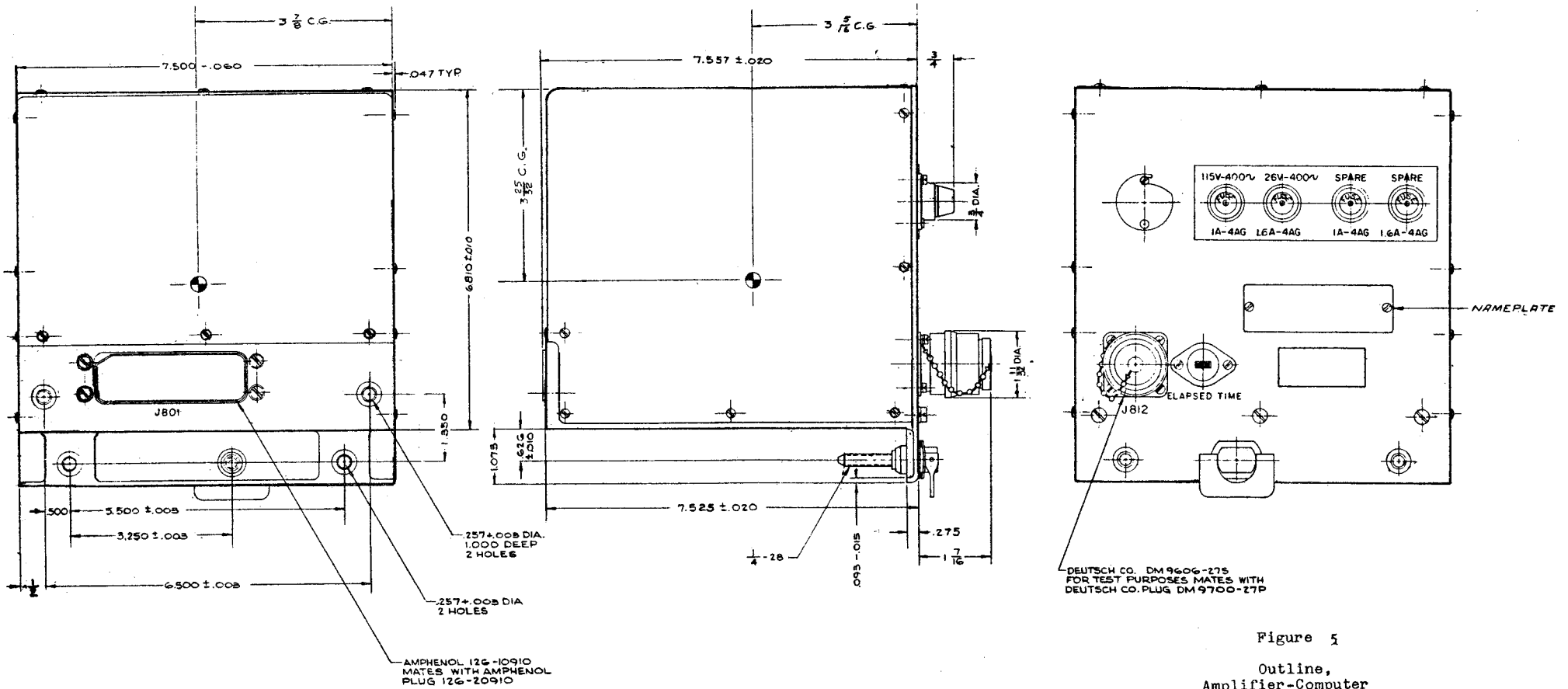


Figure 5

Outline,
Amplifier-Computer
AM-3049/ASN-39

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81169(WP) COMPUTER SET, NAVIGATIONAL AN/ASn-39(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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